

OSSP (Operating Systems And Systems Programming) - 25CS2104E

Roll NO. : 2520030437

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SEC-8

SKILL-2:

Aim: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```
PS C:\Users\Admin> wsl
admin2007@DESKTOP-MPC8TJM:/mnt/c/Users/Admin$ cd /mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2
chmod +x run.sh
./run.sh
=====
Compiling Skill-2 with GCC...
=====
Compilation successful!
Launching minishell...
=====
OSSP Skill-2: Mini Interactive Shell Loop
Type 'help' to see available built-in commands.
=====
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2$ pwd
[CHILD PID: 809] Executing 'pwd'...
/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2$ ls
[CHILD PID: 810] Executing 'ls'...
Makefile README.md include minishell run.bat run.sh src
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2$ whoami
[CHILD PID: 811] Executing 'whoami'...
admin2007
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2$ hostname
[CHILD PID: 812] Executing 'hostname'...
DESKTOP-MPC8TJM
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2$ help
```

```
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2$ ls -l
[CHILD PID: 947] Executing 'ls'...
total 28
-rwxrwxrwx 1 admin2007 admin2007 313 Aug 3 04:27 Makefile
-rwxrwxrwx 1 admin2007 admin2007 4244 Aug 3 04:27 README.md
drwxrwxrwx 1 admin2007 admin2007 4096 Aug 3 04:26 include
-rwxrwxrwx 1 admin2007 admin2007 17288 Aug 3 06:21 minishell
-rwxrwxrwx 1 admin2007 admin2007 354 Aug 3 04:34 run.bat
-rwxrwxrwx 1 admin2007 admin2007 442 Aug 3 05:37 run.sh
drwxrwxrwx 1 admin2007 admin2007 4096 Aug 3 04:27 src
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2$ ls -a
[CHILD PID: 953] Executing 'ls'...
. .. Makefile README.md include minishell run.bat run.sh src
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2$ ls -la
[CHILD PID: 954] Executing 'ls'...
total 28
drwxrwxrwx 1 admin2007 admin2007 4096 Aug 3 06:21 .
drwxrwxrwx 1 admin2007 admin2007 4096 Aug 3 04:26 ..
-rwxrwxrwx 1 admin2007 admin2007 313 Aug 3 04:27 Makefile
-rwxrwxrwx 1 admin2007 admin2007 4244 Aug 3 04:27 README.md
drwxrwxrwx 1 admin2007 admin2007 4096 Aug 3 04:26 include
-rwxrwxrwx 1 admin2007 admin2007 17288 Aug 3 06:21 minishell
-rwxrwxrwx 1 admin2007 admin2007 354 Aug 3 04:34 run.bat
-rwxrwxrwx 1 admin2007 admin2007 442 Aug 3 05:37 run.sh
drwxrwxrwx 1 admin2007 admin2007 4096 Aug 3 04:27 src
```

```
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2$ help
=====
OSSP Mini-Shell Help Guide
=====
Built-in Commands:
  cd <path>      - Change working directory
  clear          - Clear the terminal screen
  help          - Display this help message
  exit          - Terminate the interactive shell

External Commands:
  Supports executing Linux binaries (e.g. ls, pwd, cat, ps)
=====
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-2$ touch rext.txt
[CHILD PID: 966] Executing 'touch'...
```

Code:

```
#include "../include/shell.h"
#include <stdio.h>

int main() {
    char input[MAX_INPUT_SIZE];
    char *args[MAX_ARGS];
    int running = 1;

    printf("=====\n");
    printf("  OSSP Skill-2: Mini Interactive Shell Loop    \n");
    printf("  Type 'help' to see available built-in commands. \n");
    printf("=====\n");
    fflush(stdout);

    while (running) {
        shell_display_prompt();

        if (shell_read_input(input, sizeof(input)) < 0) {
            printf("\nExiting minishell. Goodbye!\n");
            break;
        }

        int num_args = shell_parse_input(input, args);
        if (num_args == 0) {
            continue;
        }

        if (!shell_execute_builtin(args, &running)) {
            shell_execute_external(args);
        }
    }
}
```